CHEM 350 Examination

Organic Chemistry: Reactions and Mechanisms – Answer Key

1. **(B)** The structure of the alkyl halide (primary, secondary, tertiary).
Primary alkyl halides favor S_N2 (backside attack possible), while tertiary halides favor S_N1 (stable carbocation intermediate). Secondary halides can proceed by either mechanism depending on conditions. Steric hindrance and carbocation stability are the key factors.

2. **(C)** Halogen atoms (e.g., $-$Cl, $-$Br).
Halogens are ortho/para-directing due to resonance donation of lone pair electrons, but they are deactivating due to their electron-withdrawing inductive effect. This makes them unique among substituents.

3. **(A)** 1-bromobutane.
Under anti-Markovnikov conditions (peroxide effect), the radical mechanism causes Br to add to the less substituted carbon (terminal position), yielding 1-bromobutane instead of the Markovnikov product 2-bromobutane.

4. **(C)** Nuclear Magnetic Resonance (NMR) spectroscopy.
NMR (especially ^{1}H and ^{13}C NMR) reveals connectivity by showing which atoms are bonded to which, through chemical shifts, coupling patterns, and integration. This makes it ideal for distinguishing constitutional isomers.

5. *Sample answer:*
Carbocation stability: Carbocations are stabilized by:

- **Hyperconjugation**: Adjacent C-H or C-C bonds donate electron density
- **Inductive effects**: Alkyl groups are electron-donating
- Stability order: tertiary > secondary > primary > methyl

Markovnikov's rule: In the addition of HX to an alkene, the hydrogen adds to the carbon with more hydrogens (less substituted), and X adds to the more substituted carbon. This occurs because:

- The mechanism proceeds through the more stable carbocation intermediate
- Addition to the more substituted position forms a more stable carbocation
- The reaction follows the pathway with the lowest energy intermediate

6. *Sample answer:*
Aldol condensation mechanism:

(a) Base abstracts an α-hydrogen from a carbonyl compound, forming an enolate ion

(b) The enolate acts as a nucleophile, attacking the carbonyl carbon of another molecule

(c) Protonation yields the aldol adduct (a β-hydroxy carbonyl compound)

(d) Under heating or acidic/basic conditions, dehydration occurs (elimination of water)

(e) This forms the α, β-unsaturated carbonyl product (conjugated system)

Conditions favoring unsaturated product:

- Elevated temperature promotes dehydration
- The α, β-unsaturated product is stabilized by conjugation
- Removal of water (Le Chatelier's principle) drives the equilibrium

7. *Sample answer:*
Kinetic control: At low temperature or short reaction time, the product formed fastest predominates (lowest activation energy path). This may not be the most stable product.

Thermodynamic control: At high temperature or with longer time (allowing equilibration), the most stable product predominates (lowest free energy).

Example - Sulfonation of naphthalene:

- At 80°C: 1-naphthalenesulfonic acid forms faster (kinetic product, less hindered position)
- At 160°C: 2-naphthalenesulfonic acid predominates (thermodynamic product, more stable due to less steric strain)
- The reaction is reversible, allowing equilibration to the more stable isomer at higher temperature

8. *Sample answer:*
Acidity increases when the conjugate base is more stable due to resonance or inductive effects:

Ethanol ($pK_a \approx 16$): The ethoxide ion (EtO^-) has limited stabilization. The negative charge is localized on oxygen with only inductive effects from the alkyl group.

Phenol ($pK_a \approx 10$): The phenoxide ion is stabilized by resonance with the aromatic ring. The negative charge delocalizes over multiple carbon atoms via resonance structures, significantly lowering the energy of the conjugate base.

Acetic acid ($pK_a \approx 4.75$): The acetate ion is stabilized by resonance between two equivalent oxygen atoms. The negative charge is equally shared between both oxygens, providing strong stabilization.

Acidity order: Acetic acid > Phenol > Ethanol. Greater resonance stabilization of the conjugate base leads to higher acidity.